

BEULAH AKINDELE

Principal Full Stack Software Engineer | Lagos, Nigeria

akibeulah@gmail.com | +234 909 142 7926 | linkedin.com/in/beulah-akindele-8093b9193 | beulahdev.onrender.com

PROFESSIONAL SUMMARY

Hands-on engineering lead in fintech and banking, currently co-managing a team of **6 engineers, 2 product managers and 2 QA** behind a Channels platform that moves interbank transfers and card, POS and USSD transactions for **500+ financial institutions across 9 African countries**, over **10+ payment gateways and switches**. Works in both stacks the platform runs on, **C#/.NET and Java/Spring Boot**, and stays close to the code: reviews designs and code, plans the roadmap and sprints, mentors and unblocks engineers, and leads the resolution of production incidents and customer escalations down to the service, queue or line responsible. Owns the full transaction lifecycle from initiation through reversals, reconfirmation and reconciliation, and designs for the moment an upstream switch fails mid-transaction: an idempotency key that rejects a retransmitted instruction rather than double-posting a credit, and a dead-letter path that surfaces a stuck transfer for reconciliation rather than losing it quietly. Applies AI where it earns its place: built the organisation's internal **RAG** documentation platform and tool-calling agents over guarded APIs, with **Claude Code and GitHub Copilot** in daily delivery. Recent results: a **96%** cut in reconfirmation queue depth, **38%** fewer failed transactions week-on-week, and institution onboarding cut from 3 days to under 5 minutes (**99.75%** faster), held at **99.9%+ uptime**.

CORE COMPETENCIES

Engineering Leadership Team Leadership, Technical Direction, Architecture & Design Reviews, Code Review & Engineering Standards, Mentoring & Coaching, Sprint Planning & Release Readiness, Technical Design Documents, Stakeholder Communication (Product, Operations, Customer Support, DevOps), Agile/Scrum

Fintech & Banking Products Payment Orchestration (Cards, POS, USSD), Card Issuing Services, Interbank Transfers, Transaction Processing, ISO 8583 & ISO 20022 Payment Messaging, Switch & Bank Integrations (10+ gateways), Wallets, Settlement & Reconciliation, Reversals & Reconfirmation, Fee Collection, KYC & Identity Verification, PCI DSS

Backend Engineering (.NET & Java) C#, .NET 8 (Core & Framework), ASP.NET Web API, Java, Spring Boot, Node.js, NestJS, Python/Django, RESTful APIs, SOAP/WCF, Background Services, Event-Driven Architecture, Distributed Systems & Microservices, System Integration

Messaging, Data & Consistency Message Queues (Azure Service Bus, Kafka, RabbitMQ, AWS SNS & SQS, BullMQ), Idempotency, Retry-with-Backoff & Circuit Breakers, Dead-Letter Reconciliation, Caching (Redis); Relational & NoSQL Databases: PostgreSQL, SQL Server, MySQL, Oracle 19C, MongoDB; schema design, query optimization, stored procedures, zero-downtime migrations

Security, Compliance & Auditability Authentication & Authorization, Role-Based & Function-Level Access Control, Maker-Checker Approval, Zero Trust, Transaction-Integrity Verification, Encrypted Credential Storage, Audit Logging & Transaction Trails, PII Log Sanitization, Secure Coding Practices

DevOps, Release & Observability CI/CD Pipelines (Jenkins, GitHub Actions, Bitbucket Pipelines, Octopus Deploy, Argo CD), Docker, Kubernetes, AWS, Azure, Monitoring & Alerting (Grafana, OpenTelemetry), Incident Response, Root-Cause Analysis, Zero-Downtime Deployment

Frontend React, Next.js (App Router), TypeScript, JavaScript (ES6+), Angular, Secure API Integration, Role-Based Dashboards

AI & Applied ML RAG Systems & LLM-Assisted Documentation, Tool-Calling Agents over Guarded APIs, Claude & OpenAI APIs, AI Coding Agents in Delivery (Claude Code, GitHub Copilot), Python, scikit-learn, TensorFlow

EXPERIENCE

Principal Full Stack Software Engineer

February 2024 – Present

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C#/.NET 8 & .NET Framework, Java/Spring Boot, Node.js, React/TypeScript, Angular, PostgreSQL, SQL Server, Oracle 19C, Redis, Azure Service Bus, Kafka, RabbitMQ, Docker, Kubernetes, Jenkins, Grafana, REST/SOAP

- **Co-lead the Channels team as a manager: 6 engineers, 2 product managers and 2 QA**, delivering a platform serving **500+ financial institutions in 9 African countries** across **10+ payment gateways and switches**: payment orchestration for card, POS and USSD channels, card issuing services, and the **full transaction lifecycle** (initiation, resolution, response handling, fee collection, reversals, reconfirmation and reconciliation).
- Run the team's delivery day to day: **roadmap and sprint planning, code review and design reviews**, mentoring and unblocking engineers, hands-on debugging, **KPI monitoring and tracking**, resolving **customer escalations**, and syncing with management on team capacity, while owning engineering standards and release readiness.
- Architected the platform's first **ISO 20022** integrations as **event-driven asynchronous services** in **C#/.NET on Azure Service Bus**: a synchronous acknowledgement returned inside the switch's SLA window while fulfilment dispatches over message queues, with **Redis-backed idempotency** keyed per message type on a **24-hour TTL** so a retransmitted instruction is rejected rather than double-posting a credit, and **exponential-backoff retry** before **dead-lettering for manual reconciliation**. Wrote it up as the **technical design document** every gateway integration has followed since, across **three upgrades and two new integrations at zero downtime**.

- Continuously run **system reviews** to spot gaps for improvement, such as the response to a production incident caused by a cross-gateway bottleneck: tracing it to shared confirmation processing and re-architecting it into **gateway-filtered batching queues** with **progressive retry throttling**, bounded parallel batch processing and **Grafana gateway-health metrics**: a **96% reduction in reconfirmation queue depth** and **38% fewer failed transactions** week-on-week at **99.9%+ uptime**.
- Own the services that settle transactions whose outcome is unclear: an always-on confirmation service driving ambiguous and timed-out interbank transfers through a **status-query and reversal state machine** across **nine gateway integrations**, and **Reconfirmation Jobs**, the **bulk reconciliation tool** operations runs after an outage, with **crash recovery** for stalled jobs and mid-flight restart and soft-stop controls. Closed a **production data-corruption incident** caused by unsynchronized session sharing across concurrent background timers.
- Designed and built **NodeSimulator**, the staging platform engineers across the organisation build and test payment integrations against: a **C#/ASP.NET** service with a React and TypeScript admin UI standing in for **ISO 8583** bank and ATM-network nodes over **raw TCP** alongside third-party REST and SOAP APIs, across **two asynchronous and six synchronous staging environments**. A JSON rule DSL, deliberate latency injection and **real-time traffic streaming** to a live dashboard reproduce **60+ upstream failure scenarios** on demand, so engineering and QA keep moving when a third-party sandbox is down.
- Own **iRestrict**, the access-control system governing which institutions and partner systems may call which banking APIs (token issue and expiry, IP allow-listing, per-service entitlement, auth audit logging). Built its **.NET 8 Backend-for-Frontend** over a legacy SOAP system so institutions cut across on **one feature switch rather than a big-bang rewrite**, added **function-level RBAC** and **maker-checker approval** on sensitive entitlement changes, and replaced a static shared-secret header with **short-lived RS256 JWTs** on a client-ID allowlist.
- Support operations and customer support in resolving customer-impacting issues without database access: built the transaction-investigation console they trace failures through and replaced an email-and-hand-written-SQL process, eliminating an average of **41 monthly engineer support requests** and cutting a 3-day cross-departmental onboarding process to under 5 minutes (**99.75% faster**). Restored an **audit trail that was silently losing rows** by diagnosing a schema constraint rejecting oversized error payloads, shipped behind a **versioned stored procedure** so database and application deploys went out independently at zero downtime.
- Reviewed platform services to drive adoption of containerization on **Kubernetes**, among other security hardening initiatives, and implemented **transaction-integrity hashes** verifying the debit and credit legs before execution, alongside **Zero Trust** inter-service authentication with short-lived tokens, encrypted credential storage and **PII log sanitization**, maintaining **PCI DSS compliance**.
- Built the organisation's **AI documentation platform**, a pipeline-triggered **RAG system** that parses doc comments on every push across all organisation repositories and generates a version-aware documentation site per project, with LLM-assisted engineering guidance over the full corpus (C#/.NET, NestJS, Angular, OpenAI API, wired into **Jenkins**, Octopus Deploy and Bitbucket).
- Modernized a **KYC verification suite across five identity endpoints**, extended it with two new offerings, and eliminated silent billing failures by decoupling billing into a dedicated worker service guarded by **retry-with-backoff and a circuit breaker**, so the request path degrades gracefully when a third-party identity provider does.
- At **Seaico Technologies Limited** (February 2024 to February 2025), led development of the Seaico Retail Platform on a **6-service Java/Spring Boot microservices architecture** with a React.js frontend, owning architecture decisions, REST API contracts, task assignment and code review across a **team of 4 developers**.
- Designed and built **TradeLink** at Seaico, a YAML configuration-driven Java middleware synchronizing trade data across Calypso, BMatch and IntegralFX at **sub-second propagation latency** over **Kafka and RabbitMQ** alongside SFTP, REST and SMTP, with dynamic field-level transformation across XML, JSON and CSV.
- Configured and tuned **Java 8 JVM runtime parameters** (heap sizing, GC settings) across Calypso Fixed Income and FX deployments at **Access Bank** (Nigeria, Ghana, Cameroon), **Zenith Bank** (Nigeria and UK) and **FCMB** on the banks' own infrastructure, and served as **technical liaison in client meetings**, translating requirements from technical and non-technical stakeholders into scoped project plans.

Full Stack & Machine Learning Engineer

October 2021 – January 2024

TBO Consulting Limited

Python, Django, React.js, Node.js, Express.js, PostgreSQL, MySQL, scikit-learn, TensorFlow

- Staffed onto a client engagement at Get Step, later renamed Buntu, where built **wallet and transfer infrastructure** integrating **Paystack and Providus** banking services for auditable, secure financial actions, then split recommendation and wallet logic into independent microservices: **450% performance improvement**, **30% throughput increase** and **35% query-execution improvement** via stored-procedure rewrites and composite indexing chosen around real query patterns.
- Architected Buntu's **two-layer personalized discovery engine**: an always-on rules-based scorer alongside an **ML ranker** fusing TF-IDF with embedding-based semantic ranking, combined by weighted score fusion and validated through **A/B rollout**, fed by an asynchronous pipeline with two-phase caching that held query-time latency low while cutting cloud compute cost.
- **Solely designed and delivered** a Talent-as-a-Service platform from Figma through cloud deployment, replacing manual hiring coordination with a self-service recruitment pipeline: **skill-based candidate matching** against open roles and an end-to-end interview-scheduling workflow integrating **Calendly and Zoom**.

PERSONAL PROJECTS

TBO VisiGuard / VMS (visiguard.org)

React 19, TypeScript, Node.js, Express, Knex/MySQL, Socket.io, Docker, Nginx, AWS EC2/RDS/S3/CloudFront/SNS/SQS, GitHub Actions

Multi-tenant visitor-management SaaS architected and deployed end-to-end: **5-tier role-based access control**, maker-checker approval on sensitive actions, real-time Socket.io visitor events, and an identity-provider sync layer unifying Azure AD and AWS IAM Identity Center behind one interface. Dockerized behind Nginx and shipped to AWS through a full **GitHub Actions CI/CD pipeline**.

Spartan / SPRTN.AI (sprtn.net)

NestJS, TypeScript, MongoDB, BullMQ + Redis, React 18, Next.js (App Router), Paystack, Flutterwave, Zilla, Claude API
Live, **revenue-generating** multi-tenant SaaS giving each tenant an isolated backend for CMS, eCommerce and real-estate listings, partitioned by companyId with per-tenant SMTP and payment configuration over a **payment aggregation layer across three gateways** (Paystack, Flutterwave, Zilla). JWT access-refresh plus API-key authentication, BullMQ/Redis-backed asynchronous jobs, and conversational agents built on a **tool-calling wrapper around Claude** that retrieves tenant-scoped data over guarded APIs.

EDUCATION

B.Eng. Electrical & Electronics Engineering, Covenant University